

Cardiovascular Pharmacology for Nursing Practice

NUR 320 — Acute Care Pharmacology • Unit 7: Cardiac Drugs

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Learning Objectives

By the end of this unit, the BSN student should be able to:

- Describe the mechanism of action of the major cardiovascular drug classes covered here.
- State the usual adult dosing range, route, and key monitoring parameters for each drug.
- Identify the highest-priority adverse effects and contraindications for NCLEX.
- Perform basic dosage and IV-rate calculations using the values provided.
- Recognize the safety "red flags" that require holding a dose or notifying the prescriber.

A central theme runs through this unit: **the margin for error in cardiac pharmacology is narrow.** Several of these drugs have a low therapeutic index, meaning the dose that helps and the dose that harms are close together. The nurse is the final safety check before a drug reaches the patient.

1. Digoxin (cardiac glycoside)

Class: Cardiac glycoside / positive inotrope.

Mechanism of action: Digoxin inhibits the sodium-potassium ATPase pump in cardiac muscle. This indirectly increases intracellular calcium, strengthening the force of myocardial contraction (positive inotropy). It also increases vagal tone, slowing conduction through the AV node and reducing heart rate (negative chronotropy).

Indications: Heart failure with reduced ejection fraction; rate control in atrial fibrillation.

Usual adult dosing:

- Oral maintenance dose: 0.125 mg to 0.25 mg once daily.
- For older adults or patients with renal impairment, 0.125 mg daily or every other day is common, because digoxin is renally cleared.
- A loading ("digitalizing") dose, when used, totals 0.75 mg to 1.25 mg given in divided doses over the first 24 hours.

Therapeutic serum level: 0.5 to 2.0 ng/mL. In heart failure, a lower target of 0.5 to 0.9 ng/mL is preferred. Levels above 2.0 ng/mL are associated with toxicity.

Monitoring:

- Apical heart rate for a full 60 seconds before every dose.
- Serum potassium — hypokalemia potentiates digoxin toxicity even when the digoxin level is "normal."
- Serum digoxin level and renal function (creatinine).

Highest-priority adverse effects: Digoxin toxicity is the headline NCLEX concern. Early signs include anorexia, nausea, vomiting, and visual disturbances — classically yellow-green halos around lights. Cardiac signs include bradycardia and a range of dysrhythmias.

Safety red flags:

- **Hold the dose and notify the prescriber if the apical heart rate is below 60 beats per minute in an adult.**
- Hypokalemia is a setup for toxicity; check the potassium before giving digoxin in a patient on diuretics.
- The antidote for severe digoxin toxicity is digoxin immune Fab.

Contraindications: Ventricular fibrillation; second- or third-degree heart block without a pacemaker.

2. Furosemide (loop diuretic)

Class: Loop diuretic.

Mechanism of action: Furosemide inhibits the sodium-potassium-chloride co-transporter in the ascending loop of Henle, producing rapid and substantial diuresis. Because it promotes excretion of potassium along with sodium and water, it is potassium-wasting.

Indications: Acute pulmonary edema, heart-failure fluid overload, edema, hypertension.

Usual adult dosing:

- Oral: 20 mg to 80 mg per dose; may be repeated or titrated upward.
- IV: 20 mg to 40 mg given slowly. **IV furosemide should be administered no faster than 20 mg per minute** to reduce the risk of ototoxicity.
- IV dosing is roughly twice as potent as the same oral dose.

Monitoring:

- Serum potassium (risk of hypokalemia), sodium, and renal function.
- Daily weight and intake/output — a 1 kg weight change roughly equals 1 liter of fluid.
- Blood pressure, watching for orthostatic hypotension.

Highest-priority adverse effects: Hypokalemia (which is especially dangerous when the patient also takes digoxin), dehydration, hypotension, and dose-related ototoxicity with rapid IV administration.

Safety red flags:

- A patient on **both furosemide and digoxin** is a classic high-risk pairing: the diuretic-induced hypokalemia raises the risk of digoxin toxicity. Monitor potassium closely.
- Hold and reassess if the patient is hypotensive or the potassium is below the normal range.

Contraindications: Anuria; known sulfonamide hypersensitivity (use caution).

3. Metoprolol (beta-1 selective beta-blocker)

Class: Beta-adrenergic blocker, cardioselective (beta-1).

Mechanism of action: Metoprolol blocks beta-1 receptors in the heart, reducing heart rate, contractility, and conduction velocity. The net effect lowers myocardial oxygen demand and blood pressure.

Indications: Hypertension, angina, post-myocardial-infarction, stable heart failure, rate control.

Usual adult dosing:

- Metoprolol tartrate (immediate-release): 25 mg to 100 mg twice daily.
- Metoprolol succinate (extended-release): 25 mg to 200 mg once daily.
- The two salt forms are **not** interchangeable milligram-for-milligram in dosing frequency — a frequent source of medication error.

Monitoring: Heart rate and blood pressure before each dose; signs of heart-failure worsening when initiating therapy.

Highest-priority adverse effects: Bradycardia, hypotension, fatigue, and masking of hypoglycemia symptoms in patients with diabetes (the tachycardia warning sign is blunted).

Safety red flags:

- **Hold and notify the prescriber if the heart rate is below 60 beats per minute or the systolic blood pressure is below 100 mmHg.**
- Do not stop a beta-blocker abruptly — abrupt withdrawal can cause rebound tachycardia, hypertension, and angina.

Contraindications: Severe bradycardia, second- or third-degree heart block, decompensated heart failure, cardiogenic shock.

4. Lisinopril (ACE inhibitor)

Class: Angiotensin-converting enzyme (ACE) inhibitor.

Mechanism of action: Lisinopril blocks the conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. The result is lower blood pressure and reduced cardiac afterload. Because aldosterone is suppressed, potassium is retained.

Indications: Hypertension, heart failure, post-MI, diabetic nephropathy protection.

Usual adult dosing: 10 mg to 40 mg once daily. Initiation is often at a lower dose (for example, 5 mg to 10 mg) with titration.

Monitoring: Blood pressure, serum potassium, and renal function (creatinine). A persistent dry cough is a common reason patients stop the drug.

Highest-priority adverse effects:

- **Angioedema** — swelling of the lips, tongue, and airway. This is a medical emergency and an absolute reason to stop the drug.
- Hyperkalemia, especially when combined with potassium-sparing diuretics or potassium supplements.
- First-dose hypotension; a persistent dry cough.

Safety red flags:

- Any facial or airway swelling after a dose is angioedema until proven otherwise — stop the drug and get help.
- **ACE inhibitors are contraindicated in pregnancy** because of fetal harm.

Contraindications: Pregnancy; history of ACE-inhibitor angioedema; bilateral renal artery stenosis.

5. Heparin (parenteral anticoagulant)

Class: Anticoagulant (unfractionated heparin).

Mechanism of action: Heparin potentiates antithrombin III, which inactivates thrombin and factor Xa, preventing the formation and extension of clots. It does not dissolve existing clots.

Indications: Venous thromboembolism treatment and prevention, acute coronary syndrome, maintaining line patency.

Usual adult dosing:

- Treatment infusions are weight-based. A common protocol uses a bolus of 80 units per kilogram followed by a continuous infusion of 18 units per kilogram per hour, then titrated to laboratory values.
- Prophylactic subcutaneous dosing is commonly 5,000 units every 8 to 12 hours.

Monitoring:

- Activated partial thromboplastin time (aPTT) for IV heparin, with a typical therapeutic target of 1.5 to 2.5 times the control value.
- Platelet count, watching for heparin-induced thrombocytopenia (HIT).
- Signs of bleeding.

Highest-priority adverse effects: Bleeding and heparin-induced thrombocytopenia.

Safety red flags:

- **The antidote for heparin is protamine sulfate.**
- A falling platelet count in a heparin patient suggests HIT — a serious reaction that requires stopping heparin.
- Heparin infusions are high-alert medications and require an independent double-check of the pump settings.

Contraindications: Active major bleeding; severe thrombocytopenia; known HIT.

6. Warfarin (oral anticoagulant)

Class: Oral anticoagulant; vitamin K antagonist.

Mechanism of action: Warfarin inhibits the synthesis of vitamin-K-dependent clotting factors (II, VII, IX, and X). Its onset is slow because circulating clotting factors must be depleted first, so heparin and warfarin are often overlapped at the start of therapy.

Indications: Long-term prevention and treatment of thromboembolism, atrial fibrillation, mechanical heart valves.

Usual adult dosing: 2 mg to 5 mg once daily initially, then adjusted to the INR.

Monitoring:

- International Normalized Ratio (INR). A typical therapeutic target is an INR of 2.0 to 3.0; mechanical valves may target higher.
- Signs of bleeding; dietary consistency of vitamin-K-rich foods (green leafy vegetables), since large swings affect the INR.

Highest-priority adverse effects: Bleeding, which becomes more likely as the INR rises above the target range.

Safety red flags:

- **The antidote for warfarin is vitamin K (phytonadione).**
- Warfarin is contraindicated in pregnancy.
- Many drugs and foods interact with warfarin; a new medication is a reason to recheck the INR.

Contraindications: Pregnancy; active bleeding; severe uncontrolled hypertension.

7. Amiodarone (antiarrhythmic)

Class: Class III antiarrhythmic.

Mechanism of action: Amiodarone prolongs the cardiac action potential and the refractory period primarily by blocking potassium channels, though it has properties spanning all four antiarrhythmic classes. The result is suppression of a wide range of atrial and ventricular dysrhythmias.

Indications: Life-threatening ventricular dysrhythmias; atrial fibrillation.

Usual adult dosing:

- Oral maintenance: 200 mg to 400 mg once daily after a loading period.
- IV in acute settings is given per ACLS or institutional protocol.

Monitoring: Amiodarone is notable for organ toxicities that develop over time. Baseline and periodic monitoring includes thyroid function (it contains iodine and can cause both hypo- and hyperthyroidism), liver function, pulmonary function (risk of pulmonary fibrosis), and eye exams (corneal deposits).

Highest-priority adverse effects: Pulmonary toxicity, thyroid dysfunction, hepatotoxicity, photosensitivity, and a blue-gray skin discoloration with long-term use.

Safety red flags: New or worsening shortness of breath or a dry cough in a patient on amiodarone may signal pulmonary toxicity and must be reported. The drug has a very long half-life, so effects and interactions persist for weeks after stopping.

Contraindications: Severe bradycardia and high-degree heart block without a pacemaker; iodine hypersensitivity.

8. Nitroglycerin (organic nitrate)

Class: Organic nitrate / vasodilator.

Mechanism of action: Nitroglycerin is converted to nitric oxide, which relaxes vascular smooth muscle. Venous dilation reduces preload and myocardial oxygen demand; at higher doses it dilates coronary arteries. The net effect relieves angina.

Indications: Acute angina, chronic angina prophylaxis, acute coronary syndrome, acute heart failure.

Usual adult dosing:

- Sublingual tablet or spray for acute angina: one dose, repeated every 5 minutes for up to 3 doses. **If chest pain persists after the first dose, the patient should call emergency services** rather than wait for all three.

- Transdermal patches are dosed once daily with a deliberate nitrate-free interval (often 10 to 12 hours) to prevent tolerance.

Monitoring: Blood pressure and the character of the chest pain; headache is an expected, common effect.

Highest-priority adverse effects: Hypotension, reflex tachycardia, and throbbing headache.

Safety red flags:

- **Nitrates are absolutely contraindicated with phosphodiesterase-5 inhibitors** (sildenafil and related erectile-dysfunction drugs) because the combination can cause profound, life-threatening hypotension. Always ask.
- Hold for systolic blood pressure below 90 mmHg.
- Sublingual tablets must be stored in their original dark glass container; they degrade with light and air.

Contraindications: Recent phosphodiesterase-5 inhibitor use; severe hypotension; increased intracranial pressure.

9. Aspirin and Clopidogrel (antiplatelet agents)

Class: Antiplatelet agents.

Mechanism of action: Aspirin irreversibly inhibits cyclooxygenase, blocking thromboxane-mediated platelet aggregation. Clopidogrel blocks the platelet ADP receptor by a different pathway. Because they act on different steps, the two are often used together ("dual antiplatelet therapy") after acute coronary syndrome or stent placement.

Indications: Prevention of arterial thrombosis in coronary artery disease, after myocardial infarction, and after coronary stenting.

Usual adult dosing:

- Aspirin: a chewable loading dose of 162 mg to 325 mg during a suspected myocardial infarction, then 81 mg daily for maintenance.
- Clopidogrel: a loading dose followed by 75 mg once daily.

Monitoring: Signs of bleeding (bruising, dark stools, prolonged bleeding from cuts); gastrointestinal upset with aspirin.

Highest-priority adverse effects: Bleeding and gastrointestinal irritation or ulceration with aspirin.

Safety red flags: Antiplatelet effects add to those of anticoagulants such as heparin and warfarin; a patient on several agents that affect clotting needs careful bleeding assessment. Aspirin is avoided in children with viral illness because of the risk of Reye syndrome.

Contraindications: Active bleeding; aspirin hypersensitivity; aspirin in children with viral illness.

10. Dosage Calculation Practice

These worked examples use only the drugs and concentrations introduced above. Show your work; the goal is a defensible method, not a memorized answer.

Example 1 — Tablet calculation. Order: digoxin 0.125 mg PO daily. Available: 0.25 mg scored tablets. Calculation: $0.125 \text{ mg} \div 0.25 \text{ mg per tablet} = 0.5 \text{ tablet}$ (one-half of a scored tablet).

Example 2 — IV volume. Order: furosemide 40 mg IV now. Available: furosemide 10 mg/mL. Calculation: $40 \text{ mg} \div 10 \text{ mg per mL} = 4 \text{ mL}$. Safety check: at a maximum rate of 20 mg per minute, this 40 mg dose should be given over at least 2 minutes.

Example 3 — Weight-based bolus. Order: heparin bolus 80 units/kg for a patient weighing 70 kg. Available: heparin 1,000 units/mL. Calculation: $80 \text{ units/kg} \times 70 \text{ kg} = 5,600 \text{ units}$. Volume = $5,600 \text{ units} \div 1,000 \text{ units per mL} = 5.6 \text{ mL}$.

Example 4 — Continuous infusion rate. Order: heparin infusion 18 units/kg/hr for the same 70 kg patient. Available: 25,000 units in 250 mL (100 units/mL). Calculation: $18 \text{ units/kg/hr} \times 70 \text{ kg} = 1,260 \text{ units/hr}$. Rate = $1,260 \text{ units/hr} \div 100 \text{ units per mL} = 12.6 \text{ mL/hr}$.

Example 5 — Safety-gated administration. Order: metoprolol tartrate 50 mg PO twice daily. The patient's apical heart rate is 54 beats per minute. Action: **Hold the dose and notify the prescriber.** A heart rate below 60 is a red flag for a beta-blocker; performing the calculation is irrelevant if the dose should not be given.

Example 6 — Reconstituted concentration. Order: lisinopril is not relevant here; instead consider amiodarone maintenance 200 mg PO daily. Available: 100 mg tablets. Calculation: $200 \text{ mg} \div 100 \text{ mg per tablet} = 2 \text{ tablets}$.

Example 7 — Titration check. A heparin infusion is running at 12.6 mL/hr (100 units/mL). The aPTT comes back subtherapeutic and the protocol calls for a rate increase to 15 mL/hr. Calculation: $15 \text{ mL/hr} \times 100 \text{ units per mL} = 1,500 \text{ units/hr}$. The nurse confirms the new rate against the protocol and performs an independent double-check before changing the high-alert pump.

Example 8 — Aspirin loading dose. Order: chewable aspirin 324 mg now for suspected myocardial infarction. Available: 81 mg chewable tablets. Calculation: $324 \text{ mg} \div 81 \text{ mg per tablet} = 4 \text{ tablets}$.

Patient Teaching Priorities

Medication safety does not end at administration; what the patient understands at discharge determines whether the drug helps or harms at home.

- **Digoxin:** Teach the patient to check their pulse and to report a pulse below 60, nausea, or visual changes such as yellow-green halos.

- **Furosemide:** Take in the morning to avoid nighttime urination; report dizziness; eat potassium-containing foods if directed.
 - **Metoprolol and other beta-blockers:** Do not stop abruptly; rise slowly to avoid orthostatic dizziness; diabetics should know that the drug can mask a racing heart as a low-blood-sugar warning.
 - **Lisinopril:** Report any facial or tongue swelling immediately; a dry cough is common but should be reported.
 - **Warfarin:** Keep vitamin-K intake consistent; report unusual bleeding or bruising; keep INR appointments; tell every provider you take it.
 - **Nitroglycerin:** Sit down before taking it; call emergency services if chest pain persists after the first dose; store tablets in the original dark container.
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11. Putting It Together: The Nurse as the Last Safety Check

Every drug in this unit can help a patient and every drug in this unit can harm a patient. Three habits prevent most errors:

1. **Assess before you administer.** Heart rate for digoxin and metoprolol; potassium for digoxin and furosemide; INR for warfarin; aPTT and platelets for heparin.
2. **Know the antidote.** Digoxin immune Fab for digoxin, protamine sulfate for heparin, vitamin K for warfarin.
3. **Respect the red flags.** A heart rate below 60, a systolic pressure below 100, facial swelling, a falling platelet count, or an INR above target are all signals to stop and think rather than proceed.

The calculation is only the last step. The clinical judgment around it is the real work.